1.1 limits → Method

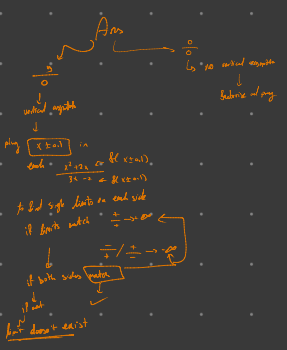
↓

exercise

1. Check $f(x)$, for some limits
↓
put checking $f(c)$ gets the right ans

Ans

$\frac{5}{0}$ → vertical asymptote → plug $\boxed{x \pm 0.1}$ in



$\frac{0}{0}$ → no vertical asymptote → factorise and prey

each $= \frac{x^2+7x}{3x-2}$ ← $f(x \pm 0.1)$, $f(x \pm 0.1)$

to find sign limits on each side

if limits match $\frac{+}{+} \to +\infty$

$\frac{-}{+} / \frac{+}{-} \to -\infty$

if both sides match

↓

if not

limit doesn't exist

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1. $\lim_{x \to 1} \boxed{x^2 + 3x - 5}$ $f(x)$

$f(1) = 1 + 3 - 5 = -1$

$\boxed{\lim_{x \to 1} f(x) = -1}$ ✓

8. $\lim_{x \to 0} \boxed{x^3 - 3x^2 + x - 5}$ $f(x)$

$f(0) = -5$

$\boxed{\lim_{x \to 0} x^3 - 3x^2 + x - 5 = -5}$ ✓

9. $\lim_{x \to 0} \boxed{\frac{x+1}{x^2+3x}}$

$f(x) = \frac{1}{0}$ ✗ → has va.

$\frac{x+1}{x(x+3)}$

$x = -0.1$

$\frac{-0.1-1}{-0.1+3} = \frac{-}{+} = \boxed{-}$

$\boxed{\lim_{x \to 0^-} f(x) = -\infty}$

One sided limit doesn't match

↓

from left $\frac{+}{- \cdot +}$

$x^- = \boxed{-}$

from right $\frac{+}{+ \cdot -} = \boxed{-}$

$x^+ = \boxed{-}$

$\boxed{x^+ \neq x^-}$

↓

$\boxed{\text{doesn't exist}}$

10. $\lim_{x \to 3} \boxed{\frac{x^2 - 2x - 3}{x^2 - 4x + 3}}$

$f(3) = \frac{3^2 - 2 \cdot 3 - 3}{3^2 - 4 \cdot 3 + 3} = \frac{9-6-3}{9-12+3} = \frac{0}{0}$ ✗

$\frac{(x-2)\cancel{(x-1)}}{(x+4)\cancel{(x-1)}}$ ✗

$g(x) = \frac{(x-2)}{(x+4)}$

$g(3) = \frac{(3-2)}{(3+4)} = \boxed{\frac{1}{7}}$

$\boxed{\lim_{x \to 3} g(x) = \frac{1}{7}}$ ✗

10. $\lim_{x \to 3} \frac{x^2 - 2x - 3}{x^2 - 4x + 3}$

$\frac{\cancel{(x-3)}(x+1)}{\cancel{(x-3)}(x-1)}$

$\frac{x+1}{x-1}$

$\frac{3+1}{3-1} = \frac{4}{2} = 2$

$\boxed{\lim_{x \to 3} f(x) = 2}$

11. $\lim_{x \to -1} \boxed{\frac{x^2 + 8x + 7}{x^2 + 6x + 5}}$

$f(-1) = \frac{(-1)^2 + 8(-1) + 7}{(-1)^2 + 6(-1) + 5} = \frac{1 - 8 + 7}{1 - 6 + 5} = \frac{0}{0}$ ✗

$\frac{(x+7)\cancel{(x+1)}}{\cancel{(x+1)}(x+5)}$

$g(x) = \frac{(x+7)}{(x+5)}$

$g(-1) = \frac{-1+7}{-1+5} = \frac{6}{4}$

$= \frac{3}{2}$

$\boxed{\lim_{x \to -1} f(x) = \frac{3}{2}}$ ✓

12. $\lim_{x \to 2} \frac{x^2 + 7x + 10}{x^2 - 4x + 4}$ → $a(1)$, → $b(x)$

$f(2) = \frac{2^2 + 7(2) + 10}{2^2 - 4(2) + 4}$

$= \frac{4 + 14 + 10}{4 - 8 + 4} = \frac{28}{0}$ → vertical asymptote

$a(1.9) = 26.91$ (+) $= +$

$b(1.9) = 0.01$ (+)

$\lim_{x \to 2^-} f(x) = +\infty$ ✓